

Load Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1155

Forest Junction 138 kV
February 20, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1155. The Interconnection Customer proposes a 87.0 MW large load (data center) facility interconnecting at the Forest Junction 138 kV substation in the Virginia area. The request was received on February 22, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 447 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$21,089,000.

3. Network Impacts

3.1 Load Deliverability Analysis

For the loss of ASPEN SUBSTATION – CRANDALL 345.0 kV Line, the DECLO – SALAMANCAROCHESTER ST 138.0 kV Line loads to 124.8% of its 287 MVA rating. Project DFAX: 10.6%.

For the loss of DAYTON – CHATOM 345.0 kV, the MOUNT CLINTON – HARRISONBURG 138.0 kV line loads to 123.9% of its 267 MVA rating. Project DFAX: 71.3%.

The FOREST JCT – MOUNT CLINTON 138.0 kV is overloaded to 122.8% of its 253 MVA emergency rating for the loss of the CHATOM – DAYTON 345.0 kV. The project contributes 71.3% DFAX (62.0 MW) to this overload.

The CARD – YUCAIPA 138.0 kV is overloaded to 121.3% of its 267 MVA emergency rating for the loss of the CRANDALL – ASPEN 345.0 kV. The project contributes 8.6% DFAX (7.5 MW) to this overload.

CARD – CRAGGY 138.0 kV: 120.7% loading (258 MVA) following the outage of the CRANDALL – ASPEN 345.0 kV; DFAX = 8.6%.

CRAGGY – STUARTS DRAFT SUBSTATION 138.0 kV: 119.0% loading (239 MVA) following the outage of the ASPEN SUBSTATION – CRANDALL 345.0 kV; DFAX = 8.6%.

VERONA – GREENBANK SUBSTATION 138.0 kV: 117.7% loading (288 MVA) following the outage of the CRANDALL – DAYTON 345.0 kV; DFAX = 7.6%.

For the loss of CRANDALL – ASPEN 345.0 kV L, the STUARTS DRAFT SUBSTATION – HARRY ALLEN 138.0 kV loads to 117.3% of its 233 MVA rating. Project DFAX: 8.6%.

The VERONA – REDLAND SUBSTATION 138.0 kV is overloaded to 116.7% of its 298 MVA emergency rating for the loss of the DAYTON – CRANDALL 345.0 kV. The project contributes 7.6% DFAX (6.6 MW) to this overload.

ERIE – FISHERSVILLE 138.0 kV: 113.9% loading (201 MVA) following the outage of the CRANDALL – ASPEN 345.0 kV; DFAX = 8.6%.

For the loss of CRANDALL – ASPEN 345.0 kV, the FISHERSVILLE – HARRY ALLEN 138.0 kV Line loads to 112.5% of its 219 MVA rating. Project DFAX: 8.6%.

N RIV D.P SUBSTATION – DOWNTOWN 138.0 kV Line: 105.2% loading (209 MVA) following the outage of the CRANDALL – DAYTON 345.0 kV L; DFAX = 18.2%.

CRANDALL – ERIE SUBSTATION 138.0 kV: 100.2% loading (204 MVA) following the outage of the ASPEN – CRANDALL 345.0 kV Line; DFAX = 8.6%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
DECLO – SALAMANCAROCHESTER ST 138.0 kV Line	Rebuild line with higher-capacity conductor	\$33,561,000	17% (\$5,654,000)
MOUNT CLINTON – HARRISONBURG 138.0 kV line	Rebuild line with higher-capacity conductor	\$4,229,000	79% (\$3,355,000)
FOREST JCT – MOUNT CLINTON 138.0 kV	Rebuild line with higher-capacity conductor	\$5,805,000	79% (\$4,605,000)
CARD – YUCAIPA 138.0 kV	Rebuild line with higher-capacity conductor	\$5,570,000	21% (\$1,155,000)
CARD – CRAGGY 138.0 kV	Rebuild line with higher-capacity conductor	\$12,327,000	21% (\$2,556,000)
CRAGGY – STUARTS DRAFT 138.0 kV	Rebuild line with higher-capacity conductor	\$4,057,000	21% (\$841,000)
VERONA – GREENBANK 138.0 kV	Rebuild line with higher-capacity conductor	\$3,267,000	7% (\$232,000)
HARRY ALLEN – STUARTS DRAFT 138.0 kV	Rebuild line with higher-capacity conductor	\$4,451,000	21% (\$923,000)
REDLAND – VERONA 138.0 kV	Rebuild line with higher-capacity conductor	\$4,621,000	7% (\$329,000)
FISHERSVILLE – ERIE 138.0 kV	Rebuild line with higher-capacity conductor	\$2,755,000	21% (\$571,000)
FISHERSVILLE – HARRY ALLEN 138.0 kV line	Rebuild line with higher-capacity conductor	\$1,674,000	21% (\$347,000)
DOWNTOWN – NORTH RIV D.P 138.0 kV	Reconductor line	\$1,811,000	19% (\$341,000)
CRANDALL – ERIE 138.0 kV Line	Reconductor line	\$869,000	21% (\$180,000)
	Total allocated to project		\$21,089,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.